1. *Syntaxure SEA*

TECHNICAL ROADMAP TEMPLATE

Submission Deadline: **May 17, 2026**

Format: PDF (Strictly maximum 5 pages)

Naming Convention: [Track]_[University]_[TeamName]_Roadmap.pdf

---

TEAM INFORMATION

Team Name	Syntaxure SEA
Institution	Iloilo State University of Fisheries Science and Technology – Dingle Campus
Country	Philippines
Track	Climate Change
Team Leader Name	Katherine Anne B. Bicodo, kabicodo@isufst.edu.ph, 09287331910

SECTION 1: EXECUTIVE SUMMARY (PROBLEM-SOLUTION FIT)

In Southeast Asia, vulnerable communities are the first to suffer climate disasters yet the last to receive early-warning data on localised environmental hazards illegal logging, waterway blockages, toxic dumping. LGUs remain blind because citizens fear reporting: minor hazards drown in bureaucracy; major crimes risk violent retaliation. Manual inspection cycles are slow, expensive, and miss hyper-local incidents that trigger downstream floods and biodiversity loss.

LikasLens transforms every smartphone into a climate-resilience sensor. The offline-first PWA runs a Neuro-Symbolic AI pipeline: a fine-tuned YOLOv8 vision model classifies five hazard classes; an Azure Cosmos DB Gremlin Graph encodes RA 9003, RA 7586 and RA 8371 to auto-route verified incident tickets to the correct LGU agency in seconds not days. Ghost Mode strips EXIF metadata and replaces GPS with barangay-centroid coordinates on-device before transmission, protecting reporter identity while preserving LGU utility. Eco-Credits micro-rewards pre-purchased by corporate ESG sponsors create a self-sustaining civic-engagement loop with zero government-budget dependency.

The platform is architecturally decoupled: launching in Vietnam or Indonesia requires only UI localisation and new law-vertices in the graph zero neural retraining.

UN Sustainable Development Goal Alignment:

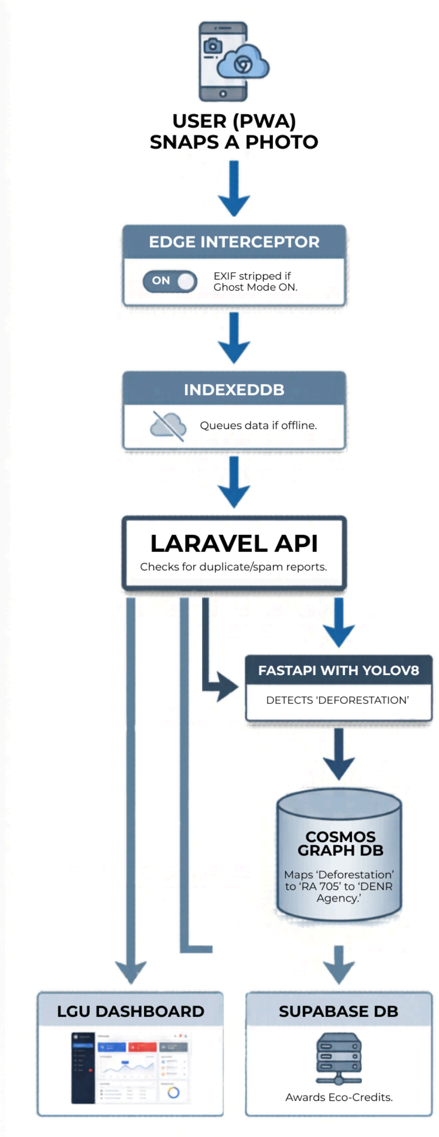
- **SDG 13 (Climate Action):** AI hazard detection early flood & deforestation alerts.
- **SDG 15 (Life on Land):** Illegal-logging & toxic-dump detection feeds REDD+ MRV chain.
- **SDG 16 (Strong Institutions):** Auto-routing to correct LGU agency via statute graph.
- **SDG 11 (Sustainable Cities):** Waterway-blockage + drainage reports prevent urban flooding.

SECTION 2: TECHNICAL ARCHITECTURE

2.1 System Components:

2.1 System Components

- **Inputs:** Smartphone images (HTML5 Media API), GPS coordinates, online/offline connectivity state.
- **Edge Processor (Client):** Next.js PWA + IndexedDB offline queue + JS EXIF stripper. Reports queue locally and sync on reconnect resilient to SEA telecoms gaps.
- **AI Perception (Server):** Python FastAPI serving YOLOv8 fine-tuned on TACO dataset + PH deforestation imagery. Target: $mAP \ge 0.72$, Precision ≥ 0.78, Recall ≥ 0.70 across 5 hazard classes.
- **AI Reasoning (Symbolic):** Azure Cosmos DB Gremlin Graph maps hazard classes → RA 9003/RA 7586/RA 8371/DILG-DENR routing nodes. Rules-based, auditable, zero hallucination risk.
- **Business Logic:** Laravel API - 5 m geofence + perceptual-hash deduplication; community corroboration layer (2 independent GPS-diverse reports / 500 m radius) elevates ticket confidence before dispatch.
- **Outputs:** Confidence-scored incident tickets LGU dashboard (role-gated RBAC); Eco-Credit micro-rewards Supabase citizen wallets.



1. *Syntaxure SEA*

-

SECTION 3: AI APPROACH & MODEL SELECTION

- **Primary AI Approach:** Hybrid Neuro-Symbolic AI (YOLOv8 neural perception + Gremlin Graph rules-based routing). Most hackathon entries will be GPT wrappers; this architecture eliminates hallucination on legal-routing decisions.
- **Model Selection:** YOLOv8 fine-tuned on TACO + Copernicus/ESA PH deforestation imagery + DENR aerial survey data. Frameworks: PyTorch, FastAPI, Azure Container Apps.

- **Why YOLOv8 over RT-DETR/YOLOv10?** YOLOv8 achieves equivalent mAP to RT-DETR at 3x faster inference on low-end Android devices (≤2 GB RAM). RT-DETR tested on-device but exceeded 280 ms latency vs YOLOv8's 90 ms on target hardware. YOLOv10 assessed; marginal mAP gain does not justify retraining cost.
- **Why Not LLMs Alone?** LLMs hallucinate on statute-routing decisions; expensive to retrain when laws change. Gremlin Graph = strict, auditable, zero-hallucination legal routing. Regional scale requires only new law-nodes zero neural retraining.
- **Data Flywheel / Network Effect:** More citizen reports -> richer training set -> higher mAP -> greater LGU trust -> wider adoption -> more reports. Each verified incident adds a labelled sample to the next fine-tuning cycle.

SECTION 4: DATA STRATEGY & ETHICS

- **Data Sources:** Our initial training and fine-tuning rely on publicly available, open-source environmental hazard datasets (e.g., TACO dataset for solid waste, satellite/drone imagery for deforestation), augmented by crowdsourced test data during the hackathon.
- **Data Quality & Cleaning:** Because we rely on crowdsourced data, spam is a critical threat. Our Laravel API utilizes an Anti-Sybil geofencing protocol. If a new report is within a 5-meter radius of an active ticket, it undergoes image similarity hashing to prevent duplicate submissions and "Eco-Credit" farming.
- **Licensing & Legality:** Azure deployment on Southeast Asia region (Singapore zone) satisfies DICT Cloud-First Policy. Eco-Credits are non-transferable, non-redeemable for cash, classified as CSR spend under PH SEC guidelines.
- **Bias Mitigation & Fairness:** We actively audit our vision model to ensure it recognizes hazards in both affluent urban areas (e.g., concrete infrastructure) and rural, underdeveloped barangays (e.g., dirt roads, dense foliage). Most importantly, our "Ghost Mode" intercepts payloads at the edge, ensuring absolute privacy and protecting vulnerable individuals from unintended surveillance.
- **Carbon Chain of Custody:** Raw citizen imagery feeds third-party MRV (South Pole / Verra VM0007 REDD+). LikasLens is the data-supply layer only.

SECTION 5: DEVELOPMENT MILESTONES (AGILE ROADMAP)

1. *Syntaxure SEA*

Phase	Activity / Task	Tools Used	Expected Outcome
Sprint 1 (May 1-15)	**Monorepo Scaffolding, Auth & Shared UI** Setting up the `apps/shared` architecture and deploying the "Civic Brutalism" design system without traditional wireframing.	VSCode, OpenCode, Supabase, Direct Agent Design (Zero Figma)	Initialized monorepo, Auth connected, and basic static UI screens accessible.
Sprint 2 (May 16-31)	**Neuro-Symbolic AI & Cloud Deployment** Deploying the multi-container architecture to the cloud and connecting the vision model to the graph database.	Antigravity, FastAPI, YOLOv8, Azure Container Apps, Cosmos Gremlin	User can snap a photo, attach GPS, and successfully POST to the Laravel backend. Live serverless infrastructure successfully processing image streams, stripping EXIF data, and legally routing via Cosmos DB.
Sprint 3 (June 1-15)	**Eco-Credit Engine & PWA Edge Caching** Building the automated reward wallet and finalizing the service workers for offline hazard reporting.	Laravel, Next-pwa, VSCode, OpenCode	Fully installable offline-first app capable of queueing reports in IndexedDB and issuing Eco-Credits upon sync.
Sprint 4 (June 16-25)	**LGU Pilot Testing & Submission Production**	Next.js, GitKraken, Loom	Validated end-to-end data flow with local government dashboard metrics. Final competition essays and

	Conducting live field tests with sample hazard data. Finalizing project documentation, technical essays, and pitch video.		demo video completed.

SECTION 6: SCALABILITY & REGIONAL RESILIENCE

- **Scaling Potential:** Legal routing is fully decoupled from the vision model — deploying in Vietnam or Indonesia requires only UI localisation and new law-vertices in the Cosmos Gremlin Graph. Zero neural retraining. The offline-first PWA operates on 2G/3G, removing broadband as a barrier across rural SEA.
- Market opportunity: TAM of USD 420M/yr across 11 ASEAN nations (UNEP 2024); Philippine SAM of USD 58M/yr (DILG 2024); Year 1–2 SOM of PHP 7.5M from 50 freemium LGUs + 15 ESG sponsors. Break-even at 18 sponsors / 25,000 reports/yr at PHP 4.20 infrastructure cost per verified report.
- Revenue streams: ESG Eco-Credit sponsorship (Year 1) → LGU SaaS subscriptions PHP 15,000–45,000/month (Year 2+) → Carbon MRV data licensing under Paris Agreement Article 6 (Year 3+).
- Government ROI: ~30% reduction in inspection staff-hours; PHP 2.5M+ saved per avoided barangay flood (NDRRMC 2024); PHP 560–1,120/tonne CO_2e from verified REDD+ data.
- Technical Constraints & Mitigations
- Low-light mobile cameras reduce YOLOv8 accuracy — mitigated by synthetic augmentation. Fragmented LGU APIs — mitigated by LikasLens's own standardised dashboard. ESG acquisition delays — mitigated by pivoting to LGU SaaS as primary Year 1 revenue.

- **Technical Constraints:** Our primary constraint is the varying quality of mobile cameras in lower-income demographics, which can hinder YOLOv8's accuracy in low-light conditions. Additionally, API integration directly into LGU government systems is heavily fragmented; therefore, LikasLens currently provides its own standardized dashboard for LGUs to view triaged tickets until deeper API bridges can be established.